UNIVERSITY OF WOLVERHAMPTON

7CS077 Dissertation

WEEK 1 – THE MASTERS DISSERTATION

Learning Outcomes

By the end of this session you will be able to:

- Describe the purpose of a Masters Dissertation
- Explain how Masters Dissertations are evaluated
- Identify the skills needed to successfully complete a master's dissertation

What is a Master's Dissertation?

At a simple level, a Master's dissertation is a long structured essay on a topic of your choice.

You will often see pointers as to what is involved in writing this 'long structured essay', such as:

- 'An independent study . . .'
- 'evidence of in-depth investigation . . .'
- 'showing clear focus . . .'
- 'demonstrating critical knowledge . . .'
- 'using appropriate research methods . . .'
- 'an awareness of ethical issues . . .'
- 'submitted no later than . . .'
- 'between 10,000–15,000 words'
- 'formatted according to . . .'

What is a Master's Dissertation?

In other words, what you are being asked do to is to conduct an independent piece of research and write it up within a particular timeframe with a limit on the number of words you can use.

To qualify as research, your work must be tackled in a systematic way.

This means that you need to identify your topic, have clearly defined objectives, explain how you intend meeting those objectives, carry out your research, analyse your findings, and conclude with reference to what you set out to do and what you have achieved. This may include making recommendations or highlighting any implications of your work.

Research typically gives new insight into a topic by addressing issues or problems or questions that you have raised or identified or asked. Research may also produce new ways of looking at, or doing, something.

What is a Master's Dissertation?

Dissertations come in all shapes, typically reflecting the type of dissertation your University department favours.

- For example, a Computer Science dissertation will be different to an Art dissertation.

Some departments require that you answer your research objectives through an extensive review of literature.

Others may require you to write a literature review and also collect your own research data (e.g. using interviews)

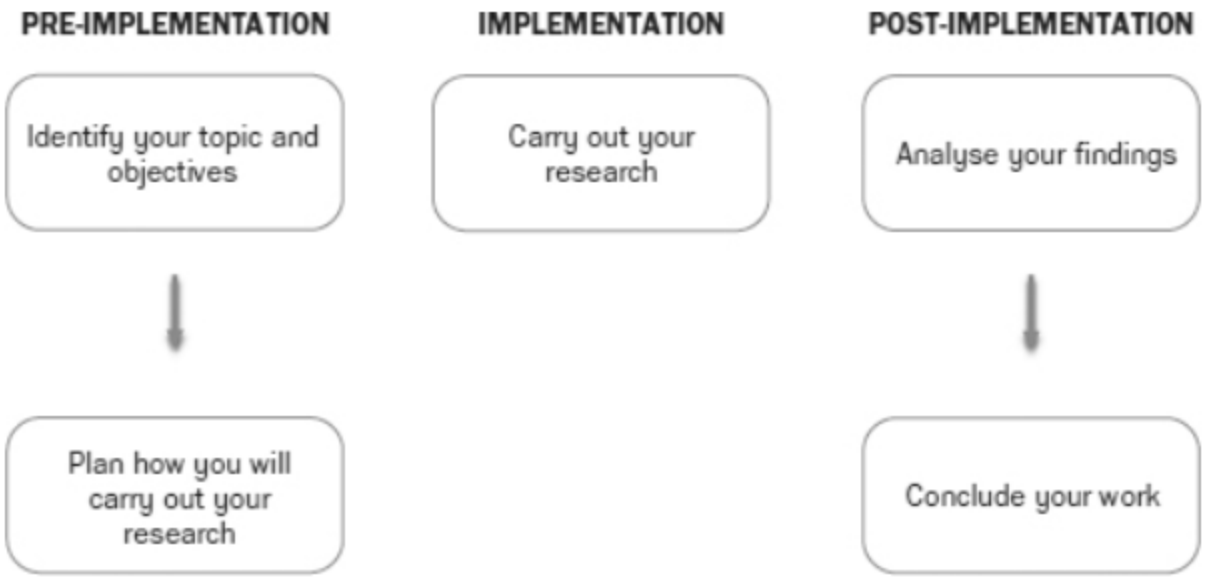
You could be required to create something (e.g. a software program), or that you do an experiment.

You will also find that academics have personal preferences, so your project supervisor may advise a dissertation structure that differs from your previous dissertations.

What is a Master's Dissertation?

Regardless of the wide range of university dissertations on offer, what all these dissertations have in common is that they require students to adopt a systematic approach i.e. that the process has all the hallmarks of competent research.

Namely, that you cover in one way or another, core research activities:



What are you being tested on?

In short, a Master's dissertation is a test to see if you can complete a research project.

Your dissertation will be submitted as a collection of chapters. Your chapters should therefore cover the stages of your research project, from start to finish.

Your evidence as a competent researcher will be expected to be in those chapters, and so it is those chapters that your markers will look at to determine your marks.

It may be that you also have to create an artefact or present your work, in which case, these other activities may also form part of the marking scheme.

Your project will conclude with a presentation in which you must present and defend your work to both your supervisor and reader.

What are you being tested on?

How you organise your chapters will depend on your research project and the dissertation submission requirements imposed by your department.

If your dissertation is theory-based and you are not collecting your own data (e.g. through interviews), then, of course, you will not require a chapter on how and why you are collecting empirical data (but you will still require a chapter on how and why you are examining prevailing theory).

Similarly, if you are not creating something, then you will not need a chapter on how and why you are creating something.

What are you being tested on?

Nonetheless, key questions that a marker will seek answers to are mainly research-related.

Typical questions include:

- Are the research objectives appropriate, focused, clear and achievable?
- Is the research placed in context and justified?
- Is the student well-read in the area and is there evidence of critical thinking?
- Are the research methods valid and reliable?
- Is there an understanding of ethical issues?
- Have the findings been described/analysed/synthesised?
- Do the conclusions follow from the findings?
- Have appropriate recommendations been made or the implications of the work understood?
- What contribution does this work make?

What are you being tested on?

From the previous list of questions, you should realise that you are not getting marks for producing earth-shattering conclusions.

That is to say, it is not your final output that is important to the marker, but what is important is:

- that your conclusions follow from your findings
- that your findings came from the proper application of your chosen research methods
- that your research methods were selected because they were appropriate to what you set out to do
- all of this is transparent to your marker

What are you being tested on?

You could have a brilliant conclusion but if your research does not show how you arrived at that conclusion, you will lose serious marks.

On the other hand, your conclusion might not lead to a guest appearance on the evening news but if the research that leads to that conclusion is:

- admirably robust
- reveals a clear understanding of the research process from start to finish

Then you are on the way to securing high marks, and rightly so.

What are you being tested on?

You can ensure transparency in your dissertation by answering what and why questions at key points in the dissertation – what you are doing and why you are doing it.

Examples of some what and why questions could be:

- What are my research objectives and why have I chosen those research objectives?
- What is the value of my work and why do I think that?
- What have others contributed in this field and why do I think that?
- What research methods am I using to do my research and why am I using those methods?

It is the why information that particularly appeals to markers: it demonstrates that you are not merely content with describing your work but that you want to explain it too.

In other words, for your work to be valued as research, process is paramount, both in terms of what you do and how you do it.

What are you being tested on?

For example, if you have used a specific value for a setting in your testing or the development of your artefact, justify why the value was used.

Imagine you trained an AI model and you used 8 epochs (iterations) in the final model because you tried a different number of epochs, both higher and lower in developmental testing.

You can explain that in your dissertation.

This will show your marker that this value wasn't "pulled out of thin air" but was actually determined through experimentation to identify what you determined to be the optimal value.

What Skills do you need?

There are two types of skills you need to complete a successful dissertation:

- 1) generic skills
- 2) skills that are research-related

The generic skills are skills that you will need irrespective of the type of dissertation that you will be doing, whether it is theory-based or contains a practical element.

As well as generic skills, you will need skills that are specific to completing a successful dissertation.

In practice, this means that you will need to know how to write the chapters – using the language of academic discourse – that typically form part of a Master's dissertation.

These chapters will chart your research journey, where the chapters collectively explain what you did, why you did it, and what you found out.

What Skills do you need?

The generic skills needed include:

Time Management Skills - Essentially, working out the time you will need to complete each part of your dissertation and when you will attempt those parts.

- If you set aside regular time-slots for your dissertation work, you will begin to view your work as a normal activity, rather than an onerous task that you keep delaying to some unspecified date in the future.

Organisational skills - You need to have a disciplined approach to your work, from structuring your dissertation, planning visits to the university library, to arranging supervisory meetings.

Communication skills - Communicating your ideas, arguments, the rationale for choices made, whether at meetings with your supervisor or in your submitted work, requires clarity of thought and expression. When communicating, in whatever form, keep it simple, keep it clear and keep it relevant.

What Skills do you need?

Listening skills - Effective communication also requires good listening skills. It is important that you learn to listen to your supervisor when advice is given.

When your supervisor suggests that perhaps you should consider altering an aspect of your work in some way, you are being told politely that it is deficient in some respect and corrective action is required. If you listen carefully, your supervisor is telling you how to gain marks and, equally important, how to avoid losing marks.

Technology skills - Information technology skills are taken for granted these days but you will need to acquire specific skills to make your dissertation work go easier. For instance, learn to navigate your university's library databases and use software to facilitate referencing. **And remember to back up your work!**

What Skills do you need?

Independent learner skills - At Master's level the learning is mainly student-centred, i.e. there is a major switch of responsibility from the tutor to you. Your supervisor acts as an adviser, albeit an important one, but you take the lead. Taking responsibility for your own work can be quite liberating but it requires the development of new skills as well as confidence in your own abilities.

Presentational skills – During a Viva, you are required to present your work to examiners. Thus you ought to be aware of how to produce a skilful presentation, including use of appropriate technologies, voice projection techniques, audience engagement tricks, etc.

- Try presenting your work periodically to friends, and your supervisor. It will provide you with invaluable experience in finding out what works and what does not work for you.

What Skills do you need?

Of course, as well as generic skills, you need research skills. Such as knowing:

- How to write a research proposal.
- How to write an abstract
- How to write an introduction that convincingly clarifies what you set out to do
- How to write a literature review (or extended contextual review), showing evidence of critical thinking on your part. Referencing is a key hallmark of academic writing
- How to write about your research methods. There are numerous research methods available to researchers. You need to be able to explain what method(s) you intend using, your data collection technique(s), and also address any ethical issues.
- How to write your findings, exhibiting skills in description, analysis, and synthesis, once again paying careful attention to the language of academic discourse.
- How to conclude your dissertation, ensuring cyclical closure.

What Qualities do you need?

In addition to generic and specific skills, there are certain personal qualities that are necessary prerequisites for the completion of a successful dissertation: self-motivation, self-confidence and self-centredness

A Master's dissertation makes demands on your time, concentration and enthusiasm over a lengthy period.

If your motivation is strong, and you adopt a positive outlook, you can face every stage with genuine curiosity, making the whole experience enjoyable and, ultimately, fruitful.

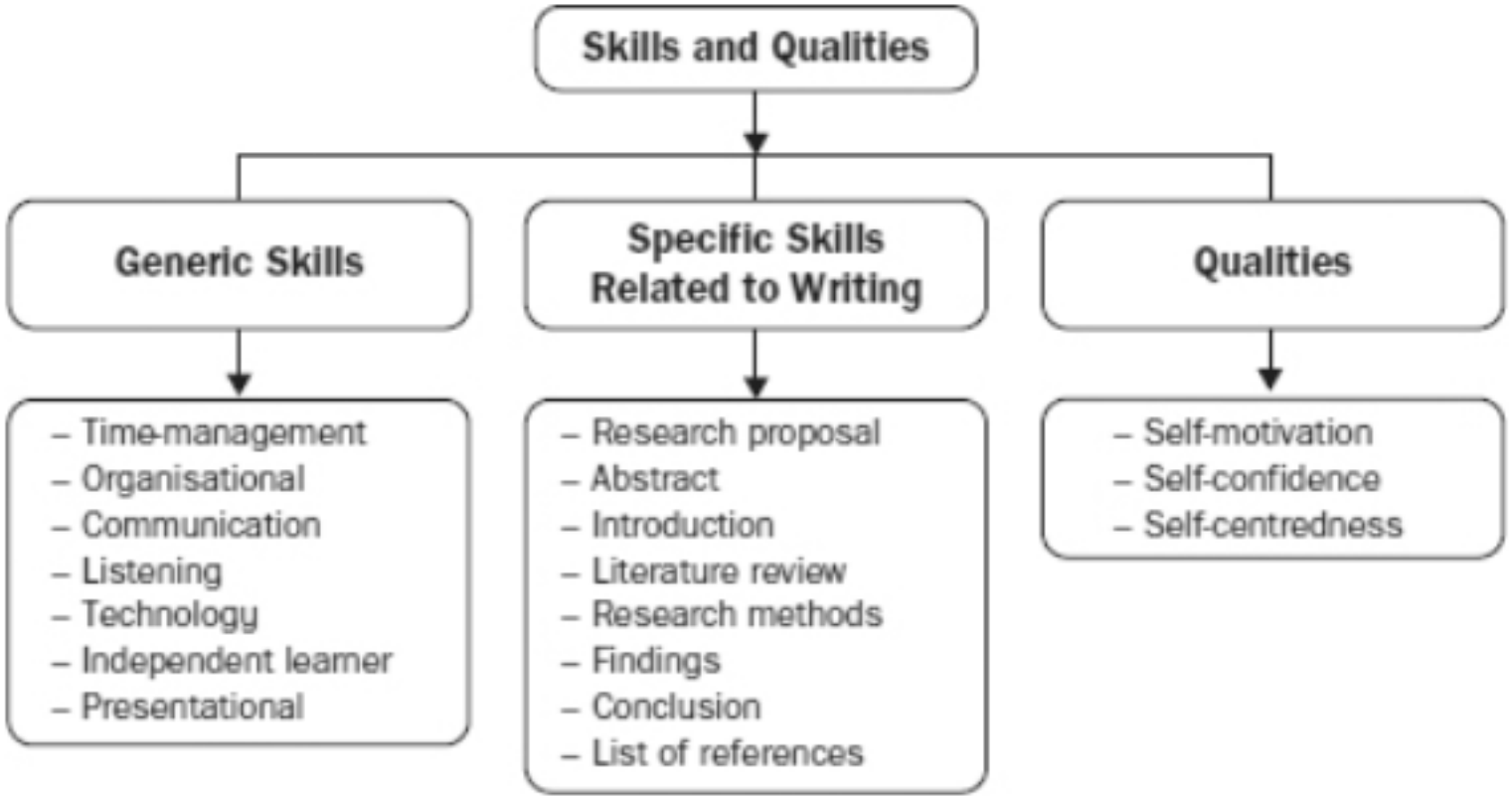
What Qualities do you need?

Have confidence in your own abilities. For example, critiquing the work of respected academics demands a level of self-confidence to allow you to express your views on what others are saying.

Your views are as valid as anyone else's, providing, of course, you back them up with supporting evidence! You are on a Master's programme because the admissions tutor has confidence that you will complete your studies. Others have faith in your abilities, so have faith in yourself.

Self-centredness is a quality that will serve you well during your dissertation. From the outset, inform those close to you that you are serious about completing your dissertation, that it is a priority for you, and that you will require their understanding and patience as you devote time and energies to give of your best.

What Skills and Qualities do you need?



Dissertation Life Cycle

The Dissertation Life Cycle highlights the conventional stages of the traditional dissertation journey:

1) produce a research proposal

2) write a Literature Review (or equivalent)

3) design appropriate research methods

4) implement empirical research

5) analyse the findings

6) conclude the research

7) submit the completed work ☺

Key Points

A Master's dissertation is an advanced study of a topic, completed independently, that adheres to the core steps expected of a research project, wherein you identify your topic and objectives, place your research in the context of what others had done, plan how you will carry out your research, carry out your research, analyse your findings, and conclude your work.

You are being tested that you can do a research project all on your own. In effect, you are being tested that you can follow a research process that is acceptable to the academic community in your subject area.

Top dissertations, at key points, answer not only what questions but also why questions.

To complete a Master's dissertation, you require a combination of generic skills, personal qualities, and skills specific to dissertation writing.

Dissertations come in all shapes and sizes, and not all dissertations reflect the traditional Dissertation Life Cycle.

For example, some may have a Systematic Review instead of the traditional Literature Review, while others may not require the student to implement their own empirical study.

Summary

You should now be able to:

- Describe the purpose of a Masters Dissertation
- Explain how Masters Dissertations are evaluated
- Identify the skills needed to successfully complete a master's dissertation

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END OF LECTURE 1



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